



## Cardiovascular Pharmacology

## Pharmacological profile of the inhibition by dihydroergotamine and methysergide on the cardioaccelerator sympathetic outflow in pithed rats

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## ABSTRACT

The present study set out to analyse the pharmacological profile of the inhibitory responses induced by the antimigraine agents dihydroergotamine (DHE) and methysergide on the tachycardic responses to preganglionic sympathetic stimulation in pithed rats. For this purpose, 132 male Wistar normotensive rats were pithed and prepared to: (i) selectively stimulate the preganglionic (C<sub>7</sub>–T<sub>1</sub>) cardiac sympathetic outflow; or (ii) receive intravenous (i.v.) bolus injections of exogenous noradrenaline. Electrical sympathetic stimulation or exogenous noradrenaline produced, respectively, frequency-dependent and dose-dependent tachycardic responses. Moreover, i.v. continuous infusions of DHE (1.8, 3.1 and 5.6 µg/kg.min) or methysergide (100, 300 and 1000 µg/kg.min) dose-dependently inhibited the tachycardic responses to sympathetic stimulation, but not those to exogenous noradrenaline. Using physiological saline or antagonists (given as i.v. bolus injections), the cardiac sympatho-inhibition induced by either DHE (3.1 µg/kg.min) or methysergide (300 µg/kg.min) was: (1) unaffected by saline (1 ml/kg); (2) partially blocked by the antagonists rauwolscine (300 µg/kg;  $\alpha_2$ ) or N-[4-methoxy-3-(4-methyl-1-piperazinyl)phenyl]-2'-methyl-4'-(5-methyl-1,2,4-oxadiazol-3-yl) [1,1'-biphenyl]-4-carboxamide hydrochloride monohydrate (GR127935, 300 µg/kg; 5-HT<sub>1B/1D</sub>); and (3) completely antagonised by the combination rauwolscine plus GR127935. These antagonists, at doses high enough to completely block their respective receptors, failed to modify the sympathetically-induced tachycardic responses *per se*. The above results, taken together, suggest that the cardiac sympatho-inhibition induced by DHE (3.1 µg/kg.min) and methysergide (300 µg/kg.min) may be mainly mediated by stimulation of both  $\alpha_2$ -adrenoceptors and 5-HT<sub>1B/1D</sub> receptors.

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## 1. Introduction

Dihydroergotamine (DHE) and methysergide have respectively been used in the acute (Saper and Silberstein, 2006) and prophylactic (Silberstein, 1998) treatment of migraine and display from moderate to high affinity for a wide array of receptors (see Table 1); stimulation of some of these receptors produces complex responses in the cardiovascular system including, among others: (i) bradycardia (Antonaccio et al., 1975; Roquebert and Grenié, 1987; Villalón et al., 1992); and (ii) inhibition of the cardiac sympathetic outflow (Sánchez-López et al., 2003, 2004; Cobos-Puc et al., 2007, 2009).

While the receptors involved in methysergide-induced cardiac sympatho-inhibition are not clearly understood (Antonaccio et al., 1975), stimulation of  $\alpha_2$ -adrenoceptors has been reported to be involved in the cardiac sympatho-inhibition induced by DHE and ergotamine (Roquebert and Grenié, 1986). More recently, it has been shown in rats that ergotamine-induced cardiac sympatho-inhibition is mediated by stimulation of  $\alpha_{2A}/\alpha_{2C}$ -adrenoceptors, D<sub>2</sub>-like receptors and, to a lesser extent, 5-HT<sub>1B/1D</sub> receptors (Cobos-Puc et al., 2009). In accordance with these findings, DHE and methysergide display reasonable affinity at  $\alpha_2$ -adrenoceptors, D<sub>2</sub>-like receptors and 5-HT<sub>1B/1D</sub> receptors (see Table 1) but, to the best of our knowledge, no study has reported whether these receptors are involved in their cardiac sympatho-inhibitory effects.

On this basis, the present study set out to analyse the receptors involved in the cardiac sympatho-inhibition induced by DHE and methysergide in pithed rats using, for this purpose, the antagonists rauwolscine ( $\alpha_2$ ), GR127935 (5-HT<sub>1B/1D</sub>) or the combination of

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**Table 1**

Affinity constants ( $pK_i$ ,  $pA_2$ ,  $IC_{50}$  nM,  $pEC_{50}$  or  $pK_D$ ) of dihydroergotamine (DHE), methysergide, GR127935 or rauwolscine for cloned human and rat receptors.

| Receptor              | DHE                                     | Methysergide        | GR127935                              | Rauwolscine      |
|-----------------------|-----------------------------------------|---------------------|---------------------------------------|------------------|
| $\alpha_2$            | 8.00 <sup>a</sup>                       | 5.59 <sup>f</sup>   | <6.00 <sup>j</sup>                    | ND               |
| h $\alpha_{2A}$       | ND                                      | ND                  | ND                                    | 8.9 <sup>l</sup> |
| h $\alpha_{2B}$       | ND                                      | ND                  | ND                                    | 8.9 <sup>l</sup> |
| h $\alpha_{2C}$       | ND                                      | ND                  | ND                                    | 9.3 <sup>l</sup> |
| $\alpha_1$            | 8.00 <sup>a</sup>                       | 5.64 <sup>f</sup>   | <6.00 <sup>j</sup>                    | ND               |
| $\alpha_{1A}$         | 8.66 <sup>b,s</sup>                     | ND                  | ND                                    | ND               |
| $\alpha_{1B}$         | 8.02 <sup>b,s</sup>                     | ND                  | ND                                    | ND               |
| $\alpha_{1D}$         | 7.87 <sup>b,s</sup>                     | ND                  | ND                                    | ND               |
| D <sub>1</sub>        | 5.32 <sup>a</sup>                       | ND                  | ND                                    | ND               |
| D <sub>2</sub>        | 8.21 <sup>a</sup> / 1.2 <sup>c,c</sup>  | 6.70 <sup>f</sup>   | <5.00 <sup>j</sup>                    | ND               |
| D <sub>3</sub>        | 6.4 <sup>c,c</sup>                      | ND                  | ND                                    | ND               |
| D <sub>4</sub>        | 8.7 <sup>c,c</sup>                      | ND                  | ND                                    | ND               |
| D <sub>5</sub>        | ND                                      | ND                  | ND                                    | ND               |
| 5-HT <sub>1A</sub>    | 9.30 <sup>a</sup>                       | 7.63 <sup>e,*</sup> | 7.20 <sup>g</sup>                     | ND               |
| h5-HT <sub>1B</sub>   | 9.22 <sup>a</sup>                       | ND                  | 9.90 <sup>j</sup> / 9.00 <sup>k</sup> | ND               |
| r5-HT <sub>1B</sub>   | 7.85 <sup>a</sup>                       | 5.82 <sup>e,*</sup> | 8.50 <sup>k</sup>                     | ND               |
| h5-HT <sub>1D</sub>   | 8.60 <sup>a</sup>                       | ND                  | 8.90 <sup>j</sup> / 8.60 <sup>k</sup> | ND               |
| 5-HT <sub>1B/1D</sub> | 7.65 <sup>a</sup>                       | 8.42 <sup>e,*</sup> | ND                                    | ND               |
| 5-HT <sub>1E</sub>    | 6.22 <sup>a</sup>                       | 6.64 <sup>g</sup>   | 6.20 <sup>j</sup> / 5.40 <sup>k</sup> | ND               |
| 5-HT <sub>1F</sub>    | 6.96 <sup>a</sup>                       | 7.47 <sup>g</sup>   | 7.30 <sup>j</sup> / 6.40 <sup>k</sup> | ND               |
| 5-HT <sub>2A</sub>    | 8.54 <sup>a</sup>                       | 8.57 <sup>e</sup>   | 7.20 <sup>j</sup> / 7.80 <sup>k</sup> | ND               |
| 5-HT <sub>2B</sub>    | 7.70 <sup>d,s</sup>                     | UA <sup>h</sup>     | 6.20 <sup>k</sup>                     | 8.3 <sup>m</sup> |
| 5-HT <sub>2C</sub>    | 7.48 <sup>e,*</sup> / 7.43 <sup>a</sup> | 8.90 <sup>i,*</sup> | 6.20 <sup>j</sup> / 7.00 <sup>k</sup> | ND               |
| 5-HT <sub>5A</sub>    | ND                                      | 7.20 <sup>i,*</sup> | 5.20 <sup>j</sup>                     | ND               |
| 5-HT <sub>5B</sub>    | ND                                      | 6.90 <sup>i,*</sup> | ND                                    | ND               |

Data taken from: <sup>a</sup>, Leysen et al. (1996); <sup>b</sup>, Görnemann et al. (2008); <sup>c</sup>, Silberstein and McCrory (2003); <sup>d</sup>, Glusa and Roos (1996); <sup>e</sup>, Hoyer (1988); <sup>f</sup>, Leysen (1985); <sup>g</sup>, Adham et al. (1993); <sup>h</sup>, Baxter et al. (1994); <sup>i</sup>, Hoyer et al. (1994); <sup>j</sup>, Pauwels (1996); <sup>k</sup>, Price et al. (1997); <sup>l</sup>, Jasper et al. (1998); <sup>m</sup>, Waincott et al. (1998). All values have been presented as  $pK_i$ , except for: <sup>s</sup>,  $pA_2$ , <sup>c</sup>,  $IC_{50}$  nM, <sup>e</sup>,  $pEC_{50}$ , <sup>\*</sup>,  $pK_D$ . Abbreviations: UA (unsurmountable antagonist). ND (not determined). *r* corresponds to rodent receptors and *h* to human receptors.

rauwolscine plus GR127935. Our results suggest that the cardiac sympatho-inhibition induced by DHE or methysergide in pithed rats may be mainly mediated by stimulation of both  $\alpha_2$ -adrenoceptors and 5-HT<sub>1B/1D</sub> receptors.

## 2. Methods

### 2.1. Animals

Male Wistar normotensive rats (250–300 g) were used in the present experiments. The animals were housed in plastic cages in a special temperature-controlled room ( $22 \pm 2$  °C, 50% humidity) on a 12:12 h light/dark cycle (with light beginning at 7 a.m.). They were allowed *ad libitum* access to food and water. All animal procedures and the protocols of the present investigation were approved by our Institutional Ethics Committee (CICUAL-Cinvestav), and followed the regulations established by the Mexican Official Norm for the Use and Welfare of Laboratory Animals (NOM-062-ZOO-1999), in accordance with the Guide for the Care and Use of Laboratory Animals in U.S.A.

### 2.2. General methods

Experiments were carried out in a total of 132 rats. After anaesthesia with diethyl ether and cannulation of the trachea, the rats were pithed by inserting a stainless-steel rod through the orbit and *foramen magnum* into the vertebral *foramen* (Shipley and Tilden 1947). The animals were artificially ventilated using an Ugo Basile pump (52 strokes/min and a stroke volume of 20 ml/kg) as previously established by Kleinman and Radford (1964). After bilateral vagosympathectomy, catheters were placed in the left and right femoral veins, for the administration of agonists and antagonists, respectively, and in the left carotid artery. The latter was connected to a Grass pressure transducer (P23 XL) for recording arterial blood pressure. Heart rate was measured with a tachograph (7P4, Grass Instrument

Co., Quincy, MA, U.S.A.) triggered from the blood pressure signal. Both blood pressure and heart rate were recorded simultaneously by a model 7 Grass polygraph (Grass Instruments Co., Quincy, MA, U.S.A.). At this point, the 132 rats were initially divided into two main groups, so that the tachycardic responses produced by preganglionic ( $C_7$ – $T_1$ ) sympathetic stimulation (group 1;  $n = 114$ ) or by i.v. bolus injections of exogenous noradrenaline (group 2;  $n = 18$ ) could be investigated under different treatments (see experimental protocols below). The tachycardic stimulus–response curves and dose–response curves elicited by electrical stimulation or exogenous noradrenaline were completed in about 60 min. Each response was elicited under unaltered values of resting heart rate. The cardiac sympathetic stimuli (0.01, 0.03, 0.1, 0.3, 1 and 3 Hz) as well as the dosing with exogenous noradrenaline (0.03, 0.1, 0.3, 1 and 3  $\mu$ g/kg) were given using a sequential schedule in 0.5 log unit increments at 3– to 5-min intervals (see below). The body temperature of each pithed rat was maintained at 37 °C by a lamp and monitored with a rectal thermometer.

### 2.3. Experimental protocols

#### 2.3.1. Protocol 1: Electrical stimulation of the cardiac sympathetic outflow

In the first group of rats ( $n = 114$ ), the pithing rod was replaced by an electrode enamelled except for 1 cm length 7 cm from the tip, so that the uncovered segment was situated at  $C_7$ – $T_1$  of the spinal cord to allow selective stimulation of the cardiac sympathetic outflow, as previously reported (Sánchez-López et al., 2003, 2004). A similar electrode was placed dorsally (Gillespie et al., 1970). Before electrical stimulation, the animals received gallamine (25 mg/kg, i.v.) to avoid electrically-induced muscular twitching. Since the cardiac sympatho-inhibitory responses to several agonists in pithed rats are particularly more pronounced at lower frequencies of stimulation (Sánchez-López et al., 2003, 2004), all the animals were systematically pretreated with desipramine (50  $\mu$ g/kg, i.v., a noradrenaline-reuptake inhibitor) 10 min before each stimulus–response curve. It should be pointed out that: (i) this dose of desipramine enhanced the tachycardic responses to sympathetic stimulation when compared to that in animals without desipramine (Villalón et al., 1999); and (ii) in animals pretreated with desipramine, the tachycardic responses to sympathetic stimulation remained unchanged after two subsequent i.v. continuous infusions of saline, indicating that there were no significant changes in the magnitude or time-course of these responses (Villalón et al., 1999). After a stable haemodynamic condition for at least 30 min, baseline values of diastolic blood pressure (a more accurate indicator of peripheral vascular resistance) and heart rate were determined. Then, the preganglionic cardiac sympathetic outflow was stimulated by applying trains of 10 s (monophasic, rectangular pulses of 2 ms duration and 50 V), at increasing frequencies of stimulation (0.01, 0.03, 0.1, 0.3, 1 and 3 Hz). When heart rate had returned to baseline levels, the next frequency was applied; this procedure was systematically performed until the stimulus–response curve had been completed (about 30 min). Then, this group of rats ( $n = 114$ ) was divided into four subgroups ( $n = 42$ , 24, 24 and 24, respectively) and new baseline values of diastolic blood pressure and heart rate were determined.

The first subgroup ( $n = 42$ ), was subdivided into seven sets ( $n = 6$  each) that received i.v. continuous infusions of, respectively: (i) 10% propylene glycol in saline (control; 0.02 ml/min; twice); (ii) DHE (0.56 and 1  $\mu$ g/kg.min); (iii) DHE (1.8 and 3.1  $\mu$ g/kg.min); (iv) DHE (5.6  $\mu$ g/kg.min); (v) methysergide (10 and 30  $\mu$ g/kg.min); (vi) methysergide (100 and 300  $\mu$ g/kg.min); and (vii) methysergide (1000  $\mu$ g/kg.min). 20 min later, the values of diastolic blood pressure and heart rate were determined again and a new stimulus–response curve was elicited during the continuous infusion.

The second subgroup ( $n = 24$ ) was subdivided into four sets ( $n = 6$  each) that received an i.v. bolus injection of, respectively: (i) saline (1 ml/kg); (ii) rauwolscine (300  $\mu$ g/kg); (iii) GR127935 (300  $\mu$ g/kg);

and (iv) the combination of rauwolscline plus GR127935. 30 min later, a stimulus–response curve was elicited again, to analyse the effects of the above compounds on the sympathetically induced tachycardic responses *per se*.

The third subgroup ( $n=24$ ) was subdivided into four sets ( $n=6$  each) that received an i.v. bolus injection of: (i) saline (control; 1 ml/kg); (ii) rauwolscline (300  $\mu\text{g/kg}$ ); (iii) GR127935 (300  $\mu\text{g/kg}$ ); and (iv) the combination of rauwolscline plus GR127935. 10 min later, the animals received an i.v. continuous infusion of DHE (3.1  $\mu\text{g/kg.min}$ ) and, 20 min later, a stimulus–response curve was elicited again during the infusion of DHE.

The fourth subgroup ( $n=24$ ) was subdivided into four sets ( $n=6$  each) that received an i.v. bolus injection of: (i) physiological saline (control; 1 ml/kg); (ii) rauwolscline (300  $\mu\text{g/kg}$ ), (iii) GR127935 (300  $\mu\text{g/kg}$ ); and (iv) the combination of rauwolscline plus GR127935. After 10 min, the animals received an i.v. continuous infusion of methysergide (300  $\mu\text{g/kg.min}$ ) and, 20 min later, a stimulus–response curve was elicited again during the infusion of methysergide.

### 2.3.2. Protocol 2: Administration of exogenous noradrenaline

The second group of rats ( $n=18$ ) was prepared as described above, but the pithing rod was left in place throughout the experiment and the administration of both gallamine and desipramine was omitted. After a stable haemodynamic condition for at least 30 min, baseline values of diastolic blood pressure and heart rate were determined. Then, the animals received i.v. bolus injections of exogenous noradrenaline (0.03, 0.1, 0.3, 1 and 3  $\mu\text{g/kg}$ ) to elicit tachycardic responses, as previously reported (Sánchez-López et al., 2004). When heart rate had returned to baseline levels, the next dose was applied; this procedure was systematically performed until the dose–response curve had been completed (about 30 min). Subsequently, this group of animals was divided into three subgroups ( $n=6$  each) that received i.v. continuous infusions of, respectively: (i) 10% propylene glycol (0.02 ml/min); (ii) DHE (3.1  $\mu\text{g/kg.min}$ ); or (iii) methysergide (300  $\mu\text{g/kg.min}$ ). 20 min later, a dose–response curve to exogenous noradrenaline was elicited again during the continuous infusion of the above compounds.

### 2.4. Drugs

Apart from the anaesthetic (diethyl ether), the compounds used in the present study (obtained from the sources indicated) were: gallamine triethiodide, desipramine hydrochloride, ( $\pm$ )-noradrenaline bitartrate, propylene glycol, rauwolscline hydrochloride and GR127935 hydrochloride (Sigma Chemical Co., St. Louis, MO, U.S.A.),

dihydroergotamine tartrate and methysergide maleate (gift from Novartis Pharma A.G., Basel, Switzerland). All compounds were dissolved in physiological saline. When needed, some drops of 10% (v/v) propylene glycol were used to dissolve DHE and methysergide, and then, the resulting solutions were finally gauged with physiological saline. This vehicle had no effect on the baseline values of diastolic blood pressure or heart rate (data not shown). GR127935 was dissolved according to the instructions of the supplier by the dispersion in distilled water to about 70 °C for 10 s and then following to cool down to room temperature. Fresh solutions were prepared for each experiment. The doses mentioned in the text refer to the salts of substances, except in the cases of DHE and methysergide, where they refer to the free base.

### 2.5. Data presentation and statistical evaluation

All data in the text and figures are presented as mean  $\pm$  s.e.mean. The baseline values and peak changes in heart rate and diastolic blood pressure produced by electrical stimulation or i.v. bolus injections of exogenous noradrenaline in the saline- and in the agonist-infused animals were determined. Using paired Student's *t*-test, the values of heart rate and diastolic blood pressure within one group or subgroup of animals were compared before (baseline values) and after: (i) i.v. administration of vehicle, gallamine, desipramine, DHE, methysergide or the antagonists; and (ii) electrical stimulation or i.v. administration of exogenous noradrenaline (peak changes). Moreover, the difference between the changes in heart rate within one group or subgroup of animals was evaluated with Student–Newman–Keuls' post-hoc test, once a two-way repeated measures analysis of variance (randomized block design) had revealed that the samples represented different populations (Steel and Torrie, 1980). Statistical significance was accepted at  $P<0.05$ .

## 3. Results

### 3.1. Systemic haemodynamic variables

Baseline values of diastolic blood pressure and heart rate in the 132 rats were  $48 \pm 1$  mmHg and  $131 \pm 2$  beats/min, respectively. These variables were not significantly modified by the i.v. bolus injection of saline, gallamine, desipramine, rauwolscline, GR127935 or the combination rauwolscline plus GR127935 (data not shown). Moreover, neither diastolic blood pressure nor heart rate were modified by the i.v. continuous infusions of 10% propylene glycol (0.02 ml/min, twice), DHE (0.56 or 1  $\mu\text{g/kg.min}$ ) or methysergide (10, 30, 100, 300 or

**Table 2**

Values of diastolic blood pressure (DBP; mmHg) and heart rate (HR; beats/min), before (baseline value) and 20 min after starting the i.v. continuous infusion of the first and second treatment with vehicle (10% propylene glycol), dihydroergotamine (DHE) or methysergide (Met).

|     | Treatment                           | Baseline value | 20 min after first treatment | 20 min after second treatment |
|-----|-------------------------------------|----------------|------------------------------|-------------------------------|
| DBP | Vehicle (0.02 ml/min; twice)        | $39 \pm 16$    | $40 \pm 19$                  | $26 \pm 7$                    |
| HR  | Vehicle (0.02 ml/min; twice)        | $130 \pm 9$    | $136 \pm 6$                  | $137 \pm 12$                  |
| DBP | DHE (0.56–1 $\mu\text{g/kg.min}$ )  | $31 \pm 4$     | $39 \pm 5$                   | $38 \pm 5$                    |
| HR  | DHE (0.56–1 $\mu\text{g/kg.min}$ )  | $125 \pm 7$    | $136 \pm 6$                  | $134 \pm 5$                   |
| DBP | DHE (1.8–3.1 $\mu\text{g/kg.min}$ ) | $40 \pm 3$     | $51 \pm 3^a$                 | $71 \pm 10^a$                 |
| HR  | DHE (1.8–3.1 $\mu\text{g/kg.min}$ ) | $119 \pm 5$    | $120 \pm 5$                  | $124 \pm 7$                   |
| DBP | DHE (5.6 $\mu\text{g/kg.min}$ )     | $43 \pm 5$     | $70 \pm 9^a$                 | –                             |
| HR  | DHE (5.6 $\mu\text{g/kg.min}$ )     | $126 \pm 5$    | $130 \pm 4$                  | –                             |
| DBP | Met (10–30 $\mu\text{g/kg.min}$ )   | $39 \pm 8$     | $45 \pm 12$                  | $36 \pm 10$                   |
| HR  | Met (10–30 $\mu\text{g/kg.min}$ )   | $138 \pm 8$    | $143 \pm 4$                  | $146 \pm 9$                   |
| DBP | Met (100–300 $\mu\text{g/kg.min}$ ) | $37 \pm 1$     | $41 \pm 3$                   | $33 \pm 4$                    |
| HR  | Met (100–300 $\mu\text{g/kg.min}$ ) | $152 \pm 5$    | $153 \pm 4$                  | $156 \pm 15$                  |
| DBP | Met (1000 $\mu\text{g/kg.min}$ )    | $43 \pm 6$     | $51 \pm 8$                   | –                             |
| HR  | Met (1000 $\mu\text{g/kg.min}$ )    | $140 \pm 4$    | $137 \pm 8$                  | –                             |

<sup>a</sup>,  $P<0.05$ ; significantly different from baseline.

It should be pointed out that two doses of each drug were systematically studied in each set of animals. Therefore, the same baseline values of diastolic blood pressure and heart rate were taken for both first and second treatments.

1000  $\mu\text{g}/\text{kg}\cdot\text{min}$ ) (see Table 2). In marked contrast, i.v. continuous infusions of DHE (1.8, 3.1 or 5.6  $\mu\text{g}/\text{kg}\cdot\text{min}$ ) produced significant increases in diastolic blood pressure which reached a steady-state condition 20 min after the infusion had been initiated, without significantly modifying heart rate (Table 2). Since two doses of DHE or methysergide were studied in each set of animals (except their highest doses which were given as a single dose; see experimental protocols), the same baseline values of diastolic blood pressure and heart rate were taken for both doses (see Table 2). It is to be noted that the increase in diastolic blood pressure produced by 3.1  $\mu\text{g}/\text{kg}\cdot\text{min}$  DHE did not significantly differ ( $P>0.05$ ) from that produced by 5.6  $\mu\text{g}/\text{kg}\cdot\text{min}$  DHE (Table 2). Furthermore, the increase of diastolic blood pressure induced by 3.1  $\mu\text{g}/\text{kg}\cdot\text{min}$  DHE ( $71 \pm 10$  mmHg) was: (i) unaffected by saline ( $69 \pm 7$  mmHg) or GR127935 ( $63 \pm 6$  mmHg); and (ii) significantly attenuated ( $P<0.05$ ) by rauwolfscine ( $50 \pm 3$  mmHg).

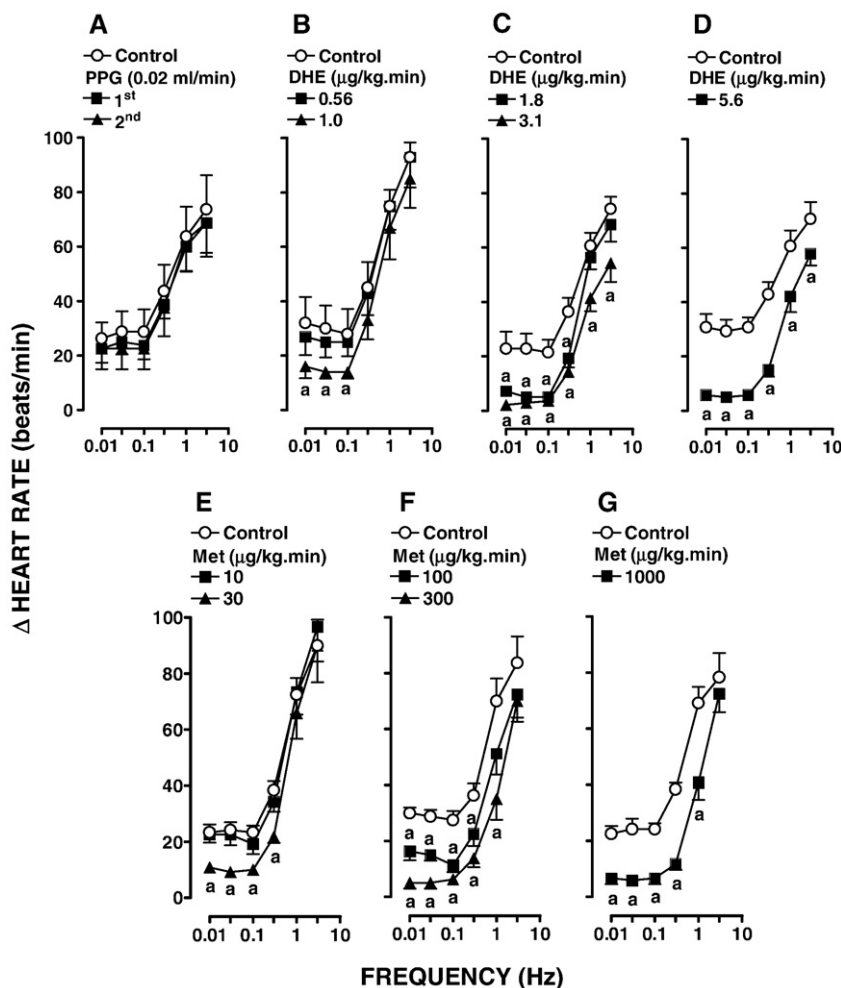
### 3.2. Effects produced by either electrical stimulation of the preganglionic cardiac sympathetic outflow or i.v. noradrenaline on heart rate

The onset of the responses produced by either stimulation of the preganglionic ( $C_7$ – $T_1$ ) cardiac sympathetic nerves (0.01–3 Hz) or i.v. bolus injections of exogenous noradrenaline (0.01–3  $\mu\text{g}/\text{kg}$ ) was immediate and resulted in, respectively, frequency-dependent and dose-dependent increases in heart rate (see below). In all cases, these increases in heart rate were statistically significant when compared to baseline values ( $P<0.05$ ). Moreover, the tachycardic responses to

exogenous noradrenaline, but not to electrical stimulation, were accompanied with vasopressor responses (data not shown), as shown by earlier studies (Villalón et al., 1999; Sánchez-López et al., 2003, 2004; Cobos-Puc et al., 2008); these vasopressor responses were not evaluated further. As previously reported, the electrically-induced tachycardic responses are due to selective cardio-stimulation since only negligible and inconsistent increases in diastolic blood pressure were observed (Cobos-Puc et al., 2007).

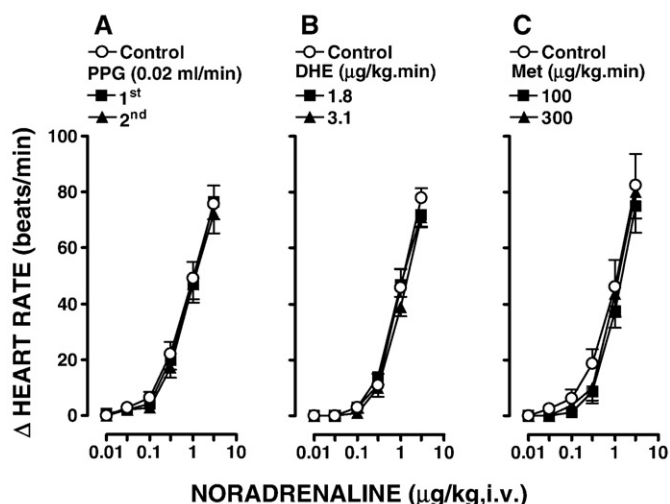
### 3.3. Effect of vehicle, DHE and methysergide on the increases in heart rate induced by either cardiac sympathetic stimulation or exogenous noradrenaline

Fig. 1 shows the tachycardic responses induced by sympathetic stimulation (0.01–3 Hz) before and during i.v. continuous infusions of: (i) propylene glycol (0.02 ml/min; twice; Fig. 1A); (ii) DHE (0.56–5.6  $\mu\text{g}/\text{kg}\cdot\text{min}$ ; Fig. 1, upper panel); and (iii) methysergide (10–1000  $\mu\text{g}/\text{kg}\cdot\text{min}$ ; Fig. 1, lower panel). The i.v. infusion of propylene glycol (Fig. 1A) or the lowest doses of either DHE (0.56  $\mu\text{g}/\text{kg}\cdot\text{min}$ ; Fig. 1B) or methysergide (10  $\mu\text{g}/\text{kg}\cdot\text{min}$ ; Fig. 1E) failed to significantly inhibit the tachycardic responses to sympathetic stimulation ( $P>0.05$ ). In contrast, 1–5.6  $\mu\text{g}/\text{kg}\cdot\text{min}$  DHE (Fig. 1B, C and D) or 30–1000  $\mu\text{g}/\text{kg}\cdot\text{min}$  methysergide (Fig. 1E, F and G) significantly and dose-dependently inhibited the sympathetically-induced tachycardic responses ( $P<0.05$ ). In contrast, the tachycardic responses to exogenous noradrenaline (see Fig. 2) remained unaltered ( $P>0.05$ )



**Fig. 1.** Increases in heart rate produced by stimulation of the cardiac sympathetic outflow before (control) and during continuous infusions of 10% propylene glycol in physiological saline (PPG) (A; 0.02 ml/min;  $n=6$ ), dihydroergotamine (DHE) (B; 0.56 and 1.0  $\mu\text{g}/\text{kg}\cdot\text{min}$ ;  $n=6$ ), (C; 1.8 and 3.1  $\mu\text{g}/\text{kg}\cdot\text{min}$ ;  $n=6$ ) and (D; 5.6  $\mu\text{g}/\text{kg}\cdot\text{min}$ ;  $n=6$ ) or methysergide (Met) (E; 10 and 30  $\mu\text{g}/\text{kg}\cdot\text{min}$ ;  $n=6$ ), (F; 100 and 300  $\mu\text{g}/\text{kg}\cdot\text{min}$ ;  $n=6$ ) and (G; 1000  $\mu\text{g}/\text{kg}\cdot\text{min}$ ;  $n=6$ ). <sup>a</sup>  $P<0.05$  vs. the corresponding control response.



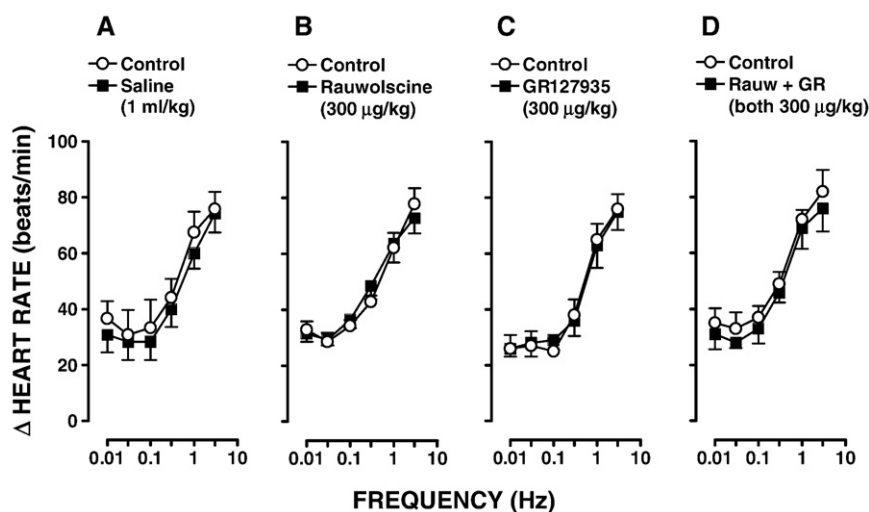


**Fig. 2.** Increases in heart rate produced by i.v. bolus injections of exogenous noradrenaline before (control) and during continuous infusions of either 10% propylene glycol in physiological saline (PPG) (A; 0.02 ml/min;  $n = 6$ ), dihydroergotamine (DHE) (B; 1.8 and 3.1  $\mu\text{g/kg min}$ ;  $n = 6$ ) or methysergide (Met) (C; 100 and 300  $\mu\text{g/kg min}$ ;  $n = 6$ ).

during i.v. continuous infusions of: (i) propylene glycol (0.02 ml/min given twice; Fig. 2A); (ii) DHE (1.8 or 3.1  $\mu\text{g/kg.min}$ ; Fig. 2B); or (iii) methysergide (100 or 300  $\mu\text{g/kg.min}$ ; Fig. 2C).

### 3.4. Effect of saline, rauwolscline or GR127935 *per se* on the sympathetically-induced tachycardic responses

Fig. 3 shows that the sympathetically-induced tachycardic responses remained unaltered after i.v. bolus injections of saline (1 ml/kg; Fig. 3A), rauwolscline (300  $\mu\text{g/kg}$ ; Fig. 3B), GR127935 (300  $\mu\text{g/kg}$ ; Fig. 3C) or the combination of rauwolscline plus GR127935 (300  $\mu\text{g/kg}$  each; Fig. 3D). Hence the above compounds were essentially devoid of any effect *per se* on the sympathetically-induced tachycardic responses. Indeed, the above doses of rauwolscline and GR127935 were high enough to completely block their respective receptors mediating sympatho-inhibition in the cardiovascular system of pithed rats (Cobos-Puc et al., 2007; Sánchez-López et al., 2004).



**Fig. 3.** Effect of i.v. bolus injections of saline (A; 1 ml/kg;  $n = 6$ ), rauwolscline (B; 300  $\mu\text{g/kg}$ ;  $n = 6$ ), GR127935 (C; 300  $\mu\text{g/kg}$ ;  $n = 6$ ) or rauwolscline plus GR127935 (Rauw + GR) (D; both 300  $\mu\text{g/kg}$ ;  $n = 6$ ) *per se* on the electrically-induced tachycardic responses.

### 3.5. Effect of saline or antagonists on the cardiac sympatho-inhibition induced by DHE and methysergide

As shown in Fig. 4, the cardiac sympatho-inhibition induced by 3.1  $\mu\text{g/kg.min}$  DHE and 300  $\mu\text{g/kg.min}$  methysergide was: (i) unaffected after saline (Fig. 4A and E); (ii) partially blocked, particularly at higher frequencies of stimulation, by rauwolscline (Fig. 4B and F) or GR127935 (Fig. 4C and G); and (ii) abolished by the combination of rauwolscline plus GR127935 (Fig. 4D and H).

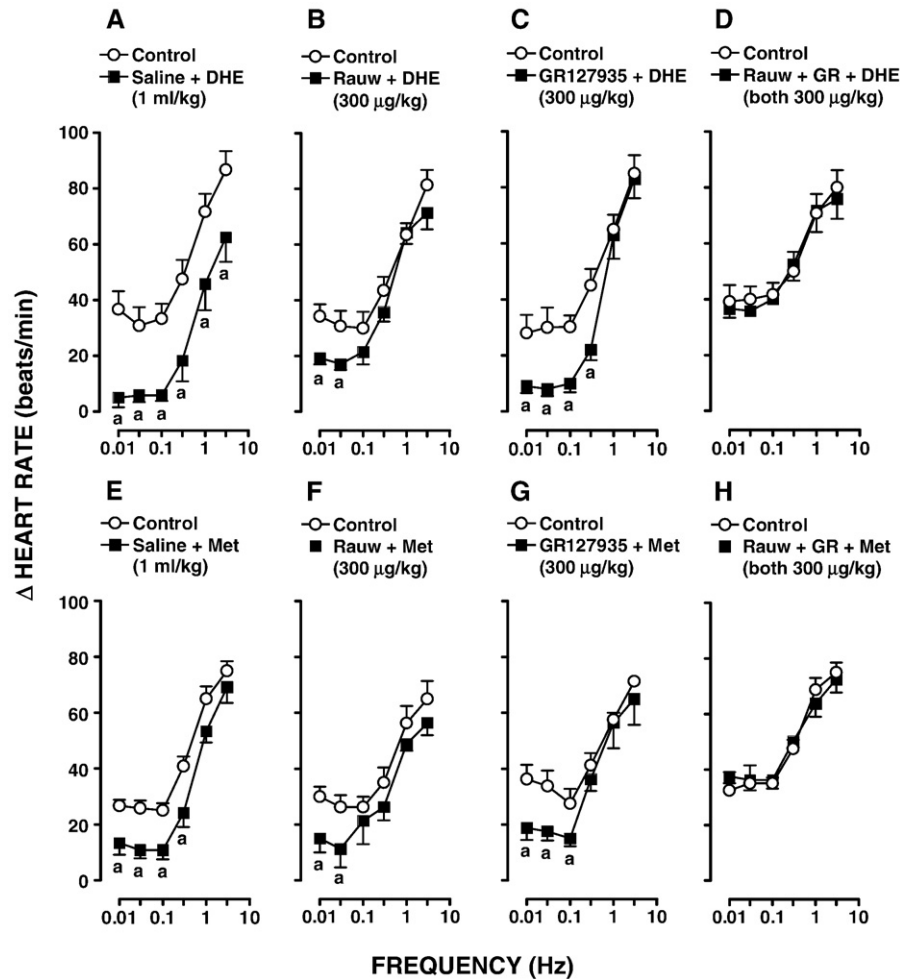
## 4. Discussion

### 4.1. General

Apart from the implications discussed below, our results show that the cardiac sympatho-inhibition induced by DHE or methysergide was: (i) unaffected by saline; (ii) partially antagonized by rauwolscline ( $\alpha_2$ ) or GR127935 (5-HT<sub>1B/1D</sub>); and (iii) abolished by the combination of rauwolscline plus GR127935. These results suggest that DHE- and methysergide-induced cardiac sympatho-inhibition may be mainly mediated by  $\alpha_2$ -adrenoceptors and 5-HT<sub>1B/1D</sub> receptors despite their complex binding profile (Table 1).

### 4.2. Systemic haemodynamic changes

Since DHE, at the doses of 1.8, 3.1 and 5.6  $\mu\text{g/kg.min}$ , elicited vasopressor responses (Table 2) and displays high affinity at  $\alpha_{1/2}$ -adrenoceptors (Table 1), the stimulation of vascular  $\alpha_{1/2}$ -adrenoceptors is implied (Roquebert and Grenié, 1986; Villalón et al., 2004; Görnemann et al., 2008). Indeed, the co-involvement of  $\alpha_2$ -adrenoceptors in these responses is reinforced by the attenuation produced by rauwolscline (see Results). In contrast, the fact that 10–1000  $\mu\text{g/kg.min}$  methysergide produced no change in diastolic blood pressure (Table 2) may be attributed to its low affinity at  $\alpha_{1/2}$ -adrenoceptors (Table 1). This suggestion is apparently inconsistent with the capability of methysergide to inhibit the cardiac sympathetic outflow by rauwolscline-sensitive mechanisms (see Fig. 4F and H). Nonetheless, one cannot categorically exclude the possibility that methysergide: (i) might display a relatively higher affinity at  $\alpha_{2C}$ -adrenoceptors (mainly involved in the prejunctional control of transmitter release from sympathetic nerves; Hein et al., 1999) than at the  $\alpha_{2A}$  and  $\alpha_{2B}$ -subtypes (mainly involved in vasoconstriction and regulation of blood



**Fig. 4.** Effects of i.v. bolus injections of saline (A, E; 1 ml/kg;  $n=6$  each set), rauwolscline (Rauw) (B, F; 300  $\mu$ g/kg;  $n=6$  each set), GR127935 (C, G; 300  $\mu$ g/kg;  $n=6$  each set) or rauwolscline plus GR127935 (Rauw + GR) (D, H; both 300  $\mu$ g/kg;  $n=6$  each set) on the inhibition of sympathetically-induced tachycardic responses elicited by a continuous infusion of dihydroergotamine (DHE) (3.1  $\mu$ g/kg.min; upper panel) or methysergide (Met) (300  $\mu$ g/kg.min; lower panel). <sup>a</sup>  $P < 0.05$  vs. the corresponding control response.

pressure, respectively; see Kanagy, 2005); and (ii) did not reach effective concentrations to produce vasopressor responses via vascular  $\alpha_1$ -adrenoceptors.

#### 4.3. Evidence that DHE and methysergide induced cardiac sympatho-inhibition by prejunctional mechanisms

Given that the inhibition produced by 3.1 and 5.6  $\mu$ g/kg.min DHE (Fig. 1C and D) or by 300 and 1000  $\mu$ g/kg.min methysergide (Fig. 1F and G) were not significantly different ( $P > 0.05$ ), the highest dose of each drug was considered to produce a supramaximal response and, accordingly, the preceding doses were chosen: (i) to analyse their effects on the tachycardic responses to exogenous noradrenaline; and (ii) for further pharmacological analysis with antagonists (see below). Hence, since DHE (Fig. 1C) and methysergide (Fig. 1F) inhibited the sympathetically-induced tachycardic responses ( $P < 0.05$ ) at the same doses that failed to affect those to exogenous noradrenaline (Fig. 2B and C;  $P > 0.05$ ), a prejunctional sympatho-inhibition is implied.

#### 4.4. Co-involvement of $\alpha_2$ -adrenoceptors and 5-HT<sub>1B/1D</sub> receptors in the cardiac sympatho-inhibition to DHE and methysergide

Our results suggest that  $\alpha_2$ -adrenoceptors and 5-HT<sub>1B/1D</sub> receptors are co-involved in the cardiac sympatho-inhibition to 3.1  $\mu$ g/kg.min DHE or methysergide 300  $\mu$ g/kg.min since both responses were partially blocked by: (i) rauwolscline (Fig. 4B and F), at a dose that

completely blocked B-HT 933-induced cardiac sympatho-inhibition (Cobos-Puc et al., 2007); or (ii) GR127935 (Fig. 4C and G) at a dose that abolished the cardiac sympatho-inhibition to the 5-HT receptor agonists CP 93,129 (5-HT<sub>1B</sub>), PNU-142633 (5-HT<sub>1D</sub>) and sumatriptan (5-HT<sub>1B/1D</sub>) (Sánchez-López et al., 2004). The above suggestion gains weight when considering that the cardiac sympatho-inhibition by: (i) DHE and methysergide was abolished by the combination rauwolscline plus GR127935 (Fig. 4D and H); (ii) DHE was significantly blocked (but not abolished) by the classic  $\alpha_2$ -adrenoceptor antagonist yohimbine (Roquebert and Grenié, 1986); (iii) ergotamine was partly blocked by rauwolscline or GR127935, but markedly blocked when both drugs were given in combination (Cobos-Puc et al., 2009). These results are consistent with other findings showing: (i) the affinity that DHE and methysergide display at  $\alpha_2$ -adrenoceptors and 5-HT<sub>1B/1D</sub> receptors (Table 1); (ii) that  $\alpha_2$ -autoreceptors play a key role in regulating noradrenaline release from sympathetic nerves (Altman et al., 1999; Trendelenburg et al., 2001); and (iii) molecular studies showing the localization of mRNA that encodes 5-HT<sub>1B/1D</sub> receptors in sympathetic nerves (Molderings et al., 2006).

Admittedly, additional experiments, which fall beyond the scope of this investigation, will be required to further identify the specific  $\alpha_2$ -adrenoceptor subtypes involved in the cardiac sympatho-inhibition by DHE and methysergide in our experiments. In this respect, previous studies analysing cardiac sympatho-inhibition in pithed rats have shown the role of the  $\alpha_{2A/2C}$ -adrenoceptor subtypes

playing a partial (when using ergotamine; Cobos-Puc et al., 2009) or full (when using B-HT 933; Cobos-Puc et al., 2007) involvement.

#### 4.5. Are other receptors involved in the cardiac sympatho-inhibition to DHE and methysergide?

The fact that DHE and methysergide display affinity for receptors other than  $\alpha_2$ -adrenoceptors and 5-HT<sub>1B/1D</sub> receptors (Table 1) leaves open the possible involvement of additional receptors including D<sub>2</sub>-like, 5-HT<sub>1A</sub> and 5-HT<sub>5A/5B</sub> receptors, which have been shown to inhibit the cardiac sympathetic outflow (Villalón et al., 1998; Sánchez-López et al., 2003; Cobos-Puc et al., 2009). However, this possibility seems unlikely since the sympatho-inhibition produced by 3.1 µg/kg.min DHE and 300 µg/kg.min methysergide was abolished by the combination rauwolscine plus GR127935 (Fig. 4D and H), which are selective for blocking their respective receptors (Table 1). Nevertheless, we cannot affirm the extent to which  $\alpha_2$ -adrenoceptors and 5-HT<sub>1B/1D</sub> receptors are involved.

In contrast with the above conclusion, it is worth noting that ergotamine induced inhibition of the cardiac sympathetic outflow in pithed rats mainly by activation of  $\alpha_{2A}/\alpha_{2C}$ -adrenoceptors, D<sub>2</sub>-like receptors and, to a lesser extent, by 5-HT<sub>1B/1D</sub> receptors (Cobos-Puc et al., 2009). The most likely explanation for this difference could reside in the binding profile and other pharmacological properties of ergotamine.

In conclusion, the cardiac sympatho-inhibition induced by 3.1 µg/kg.min DHE or 300 µg/kg.min methysergide in pithed rats may be mainly mediated by stimulation of  $\alpha_2$ -adrenoceptors and 5-HT<sub>1B/1D</sub> receptors.

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